```
# How NLTK and spaCy Work Together in My Chatbot

#The task asked me to use **both** NLTK and spaCy, not build two separate chatbots.
#Each one does a different job in the same pipeline, one after the other:

#1. **NLTK cleans the sentence.**
   #It splits the user's question into individual words,
  #then removes filler words like "the", "is", and "a".

#2. **spaCy simplifies the words.**
   #It takes those cleaned words and reduces each one to its root form.
   #For example, "studying", "studies", and "studied" all become "study".

#3. **The bot compares and responds.**
 #  Once the words are cleaned and simplified, the bot compares them
  # against a list of known topics and picks the response that matches best.

#In short:** NLTK tidies up the sentence, spaCy standardizes the words,
#and together they help the bot understand questions even when people
#phrase them differently.
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!pip install nltk spacy
!python -m spacy download en_core_web_sm
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Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-an
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✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
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```
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import nltk
import spacy
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

nlp = spacy.load('en_core_web_sm')
stop_words = set(stopwords.words('english'))
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
knowledge = {
    "when_user_says_hello": {
        "keywords": ["hello", "hi", "hey", "morning", "afternoon"],
        "response": "Hello! How can I help you today?"
    },
    "when_user_asks_about_course": {
        "keywords": ["course", "learn", "study", "subject", "class"],
        "response": "We offer AI and Machine Learning at beginner, intermediate and advanced levels."
    },
    "when_user_asks_about_fee": {
        "keywords": ["fee", "price", "cost", "pay", "payment"],
        "response": "Please contact the admin office for current fees and payment plans."
    },
    "when_user_says_bye": {
        "keywords": ["bye", "goodbye", "exit", "quit"],
        "response": "Goodbye! Have a great day."
    },
}
```

```
def preprocess(text):
    tokens = word_tokenize(text.lower())
    tokens = [t for t in tokens if t.isalpha() and t not in stop_words]
    doc = nlp(" ".join(tokens))
    return {tok.lemma_ for tok in doc}

for intent in knowledge.values():
    intent["lemmas"] = preprocess(" ".join(intent["keywords"]))

def get_response(user_input):
    user_words = preprocess(user_input)
    best_intent, best_score = None, 0
    for name, intent in knowledge.items():
        score = len(user_words & intent["lemmas"])
        if score > best_score:
            best_intent, best_score = name, score
    if best_intent:
        return knowledge[best_intent]["response"], best_intent
    return "Sorry, I didn't understand that. Can you rephrase?", None
```

```
print("Bot: Hi! Ask me a question (type 'bye' to exit).")
while True:
    user = input("You: ")
    reply, intent = get_response(user)
    print("Bot:", reply)
    if intent == "when_user_says_bye":
        break
```

```
Bot: Hi! Ask me a question (type 'bye' to exit).
You: hello
Bot: Hello! How can I help you today?
You: how much is the fee
Bot: Please contact the admin office for current fees and payment plans.
```

```
You: okay goodbye
Bot: Goodbye! Have a great day.
```